

5 There are several different human blood group systems.

Each blood group system is based on the presence or absence of certain antigens attached to the cell surface membranes of red blood cells.

In the ABO blood group system, the *ABO* gene, known as gene *I*, has three alleles. The gene is located on an autosome.

- Allele **I^A** codes for an enzyme that adds antigen A to the cell surface membrane.
- Allele **I^B** codes for an enzyme that adds antigen B to the cell surface membrane.
- Allele **I^O** codes for a non-functional enzyme, so no antigen is added.
- **I^O** is recessive to **I^A** and **I^B**.
- **I^A** and **I^B** are codominant.

(a) Explain what is meant by the term recessive **and** the term codominant.

recessive

.....

codominant

.....

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(b) Complete Table 5.1 to show the possible genotypes of the blood groups in the ABO system.

Table 5.1

blood group	genotypes
A	
B	
AB	
O	

[3]

(c) In the Rhesus blood group system, a gene codes for a protein known as the Rhesus factor. This protein is an antigen found on the surface of red blood cells.

- The dominant allele **D** codes for the Rhesus factor.
- The recessive allele **d** results in no Rhesus factor.
- A person who has the Rhesus factor is described as Rhesus positive (Rh+).
- A person who lacks the Rhesus factor is described as Rhesus negative (Rh-).
- The gene coding for the Rhesus factor is on a different autosome from the *ABO* gene.

Complete the Punnett square in Figure 5.1 to show the possible blood groups of children from these parents.

- Parent 1 is heterozygous for blood group B and heterozygous for the Rhesus factor.
- Parent 2 has blood group O and is heterozygous for the Rhesus factor.

parental phenotypes: **B Rh+** **O Rh+**

parental genotypes:

parent 1 gametes

parent 2 gametes

ratio of offspring phenotypes:

Figure 5.1

[5]

[Total: 10]